

CODE FOR FIGHTER JET GAME USING OPEN GL

INPUT CODE OR PROGRAM :

```
#include <GL/glut.h>
#include <cmath>

float jetX = 0.0f, jetY = -0.5f;
float jetAngle = 0.0f;
bool isShooting = false;
float bulletX = 0.0f, bulletY = 0.0f;

void init() {
    glClearColor(0.1f, 0.1f, 0.3f, 1.0f);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-1, 1, -1, 1);
}

void drawJet() {
    glPushMatrix();
    glTranslatef(jetX, jetY, 0.0f);
    glRotatef(jetAngle, 0.0f, 0.0f, 1.0f);
    glBegin(GL_TRIANGLES);
    glColor3f(0.8f, 0.1f, 0.1f);
    glVertex2f(0.0f, 0.1f);
    glVertex2f(-0.05f, -0.05f);
    glVertex2f(0.05f, -0.05f);
    glEnd();
    glPopMatrix();
}

void drawBullet() {
    if (isShooting) {
        glPushMatrix();
        glTranslatef(bulletX, bulletY, 0.0f);
        glColor3f(1.0f, 1.0f, 0.0f);
        glBegin(GL_QUADS);
        glVertex2f(-0.01f, -0.01f);
        glVertex2f(0.01f, -0.01f);
        glVertex2f(0.01f, 0.01f);
        glVertex2f(-0.01f, 0.01f);
        glEnd();
        glPopMatrix();
    }
}
```

```

    }
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);
    drawJet();
    drawBullet();
    glutSwapBuffers();
}

void update(int value) {
    if (isShooting) {
        bulletY += 0.02f;
        if (bulletY > 1.0f) {
            isShooting = false;
        }
    }
    glutPostRedisplay();
    glutTimerFunc(16, update, 0);
}

void handleKeys(unsigned char key, int x, int y) {
    switch (key) {
        case 'w':
            jetX += 0.05f * sin(jetAngle * M_PI / 180.0f);
            jetY += 0.05f * cos(jetAngle * M_PI / 180.0f);
            break;
        case 'a':
            jetAngle += 5.0f;
            break;
        case 'd':
            jetAngle -= 5.0f;
            break;
        case ' ':
            if (!isShooting) {
                isShooting = true;
                bulletX = jetX + 0.1f * sin(jetAngle * M_PI / 180.0f);
                bulletY = jetY + 0.1f * cos(jetAngle * M_PI / 180.0f);
            }
            break;
    }
}

int main(int argc, char** argv) {

```

```
glutInit(&argc, argv);  
glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);  
glutInitWindowSize(600, 600);  
glutCreateWindow("Fighter Jet Simulation");  
init();  
glutDisplayFunc(display);  
glutKeyboardFunc(handleKeys);  
glutTimerFunc(16, update, 0);  
glutMainLoop();  
return 0;
```

OUTPUT :



